

Chapter 2.

$$\begin{aligned} 2.1. (a) P\{Z < -0.66\} &= P\{Z > 0.66\} \\ &= 1 - P\{Z < 0.66\} \\ &= 1 - \Phi(0.66) \\ &= 1 - 0.7454 = 0.2546 \end{aligned}$$

$$\begin{aligned} (b) P\{|Z| < 1.64\} &= P\{Z < 1.64\} - P\{Z < -1.64\} \\ &\quad \downarrow \\ &\quad -1.64 < Z < 1.64 = P\{Z < 1.64\} - (1 - P\{Z < 1.64\}) \\ &= 2\Phi(1.64) - 1 \\ &= 2 \times 0.9495 - 1 \\ &= 0.8990 \end{aligned}$$

$$\begin{aligned} (c) P\{|Z| > 2.20\} &= 2P\{Z > 2.20\} \\ &\quad \downarrow \\ &\quad Z > 2.20 \text{ or } Z < -2.20 = 2(1 - P\{Z < 2.20\}) \\ &= 2(1 - \Phi(2.20)) \\ &= 2(1 - 0.9861) \\ &= 0.0278 \end{aligned}$$

2.2. $X = 2$.

2.3. By symmetry of Z ,

$$\begin{aligned} P(|Z| > X) &= P\{Z > X \text{ or } Z < -X\} \\ &= \underline{P\{Z > X\}} + \underline{P\{Z < -X\}} \\ &= 2P\{Z > X\} \end{aligned}$$

2.4. X — mean μ , variance σ^2

$$\begin{aligned} Y &= a + bX \text{ — mean } a + b\mu \\ &\quad \text{Variance } b^2\sigma^2 \end{aligned}$$

$$\Rightarrow \begin{cases} a + b\mu = \mu \\ \sigma^2 = b^2\sigma^2 \end{cases}$$

$$\Rightarrow \begin{cases} a = z\mu \\ b = -1 \end{cases}$$

$$\begin{aligned} \text{Cov}(X, Y) &= \text{Cov}(X, z\mu - X) \\ &= -\text{Cov}(X, X) \\ &= -\text{Var}(X) = -\sigma^2 \end{aligned}$$

2.6. Let X_1 and X_2 denote the life of the first and second batteries respectively.

(a). $X_1 + X_2$ is a normal random variable with mean 800 and variance

$2 \cdot 50^2$. Thus

$$\begin{aligned} P\{X_1 + X_2 > 760\} &= P\left\{\frac{X_1 + X_2 - 800}{50\sqrt{2}} > \frac{760 - 800}{50\sqrt{2}}\right\} \\ &= P\{Z > -0.5657\} \\ &= P\{Z < 0.5657\} \\ &= \Phi(0.5657) \approx 0.7142 \end{aligned}$$

(b). $X_2 - X_1$ is a normal random variable with mean 0 and variance

$2 \cdot 50^2$

$$\begin{aligned} P\{X_2 - X_1 > 25\} &= P\left\{Z > \frac{25 - 0}{50\sqrt{2}}\right\} \\ &= P\{Z > 0.3536\} \\ &= 1 - P\{Z < 0.3536\} \\ &= 1 - \Phi(0.3536) \\ &\approx 1 - 0.6382 = 0.3618 \end{aligned}$$

$$\begin{aligned} (c) P\{|X_1 - X_2| > 25\} &= 2P\{X_2 - X_1 > 25\} \\ &\approx 0.3618 \times 2 \\ &= 0.7236 \end{aligned}$$

2.7. Let X_i be the time to develop the i th print.

$$S = \sum_{i=1}^{100} X_i$$

By CLT, S approximately has a normal distribution with mean $18 \times 10 = 1800$ and standard deviation $\sqrt{100 \cdot 1^2} = \sqrt{100} = 10$.

Thus

$$\begin{aligned} (a) P\{S > 1710\} &= P\left\{Z > \frac{1710 - 1800}{10}\right\} \\ &= P\{Z > -9\} = P\{Z < 9\} \\ &= \Phi(9) \approx 1. \end{aligned}$$

$$\begin{aligned}
 (b) P\{1690 < S < 1710\} &= P\{S < 1710\} - P\{S < 1690\} \\
 &= P\{Z > -9\} - P\{Z > -11\} \\
 &= \Phi(9) - \Phi(11) \\
 &= 0.
 \end{aligned}$$

2.8. Let X_i be the mileage for flier i .

$$S = \sum_{i=1}^{30} X_i$$

By CLT, S is a normal distribution with mean $25,000 \times 30 = 750,000$ and standard deviation $\sqrt{12000^2 \times 30} = 12000\sqrt{30}$, thus

$$\begin{aligned}
 (a) P\left\{\frac{S}{30} > 25000\right\} &= P\{S > 750,000\} \\
 &= P\left\{Z > \frac{750000 - 750000}{12000\sqrt{30}}\right\} \\
 &= P\{Z > 0\} = P\{Z < 0\} \\
 &= \Phi(0) = 0.5
 \end{aligned}$$

$$\begin{aligned}
 (b) P\left\{23000 < \frac{S}{30} < 27000\right\} \\
 &= P\{S < 27000 \cdot 30\} - P\{S < 23000 \cdot 30\} \\
 &= P\left\{Z < \frac{810000 - 750000}{12000\sqrt{30}}\right\} - P\left\{Z < \frac{690000 - 750000}{12000\sqrt{30}}\right\} \\
 &= P\left\{Z < \frac{5}{\sqrt{30}}\right\} - P\left\{Z < -\frac{5}{\sqrt{30}}\right\} \\
 &= 2\Phi\left(\frac{5}{\sqrt{30}}\right) - 1 \\
 &\approx 2 \cdot 0.8194 - 1 = 0.6388.
 \end{aligned}$$

2.9. Let S_i be the stock's price after i time period, then

$$X_i = \frac{S_i}{S_{i-1}} = \begin{cases} u & \text{with probability } p \\ d & \text{with probability } 1-p \end{cases}$$

$$\begin{aligned}
 P\left\{\frac{S_{1000}}{S_0} > 1.3\right\} &= P\{X_1 X_2 \cdots X_{1000} > 1.3\} \\
 &= P\{\log(X_1) + \log(X_2) + \cdots + \log(X_{1000}) > \log 1.3\}
 \end{aligned}$$

$\log X_i$ is a random variable with mean

$$E[\log X_i] = (\log u)p + (\log d)(1-p)$$

$$= (\log 1.012) \cdot 0.52 + (\log 0.99) \cdot (1-0.52) \approx 0.0013787$$

and variance

$$\begin{aligned}
 \text{Var}(\log X_i) &= E[(\log X_i)^2] - (E[\log X_i])^2 \\
 &= p(\log u)^2 + (1-p)(\log d)^2 - 0.0013787^2 \\
 &= 0.0001206
 \end{aligned}$$

thus by CLT, $\sum_{i=1}^{1000} \log(X_i)$ is approximately

a normal random variable with mean 1.3787 and variance 0.1206.

then we have

$$\begin{aligned}
 P\left\{\sum_{i=1}^{1000} \log X_i > \log 1.3\right\} \\
 &\approx P\left\{Z > \frac{\log 1.3 - 1.3787}{\sqrt{0.1206}}\right\} \\
 &= P\{Z > -3.2146\} = P\{Z < 3.2146\} \\
 &= \Phi(3.2146) \\
 &\approx 0.9993
 \end{aligned}$$

2.10. Let X_i be the movement in period i .

$$\begin{aligned}
 E(X_i) &= (-1)(0.39) + (0)(0.20) + (1)(0.41) \\
 &= 0.02
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}(X_i) &= (-1-0.02)^2 \cdot 0.39 + (0-0.02)^2 \cdot 0.20 \\
 &\quad + (1-0.02)^2 \cdot 0.41 \approx 0.7996
 \end{aligned}$$

Let $S = \sum_{i=1}^{700} X_i$. By CLT, S is approximately

a normal distribution with mean

$$0.02 \cdot 700 = 14 \text{ and variance}$$

$$0.7996 \cdot 700 = 559.72. \text{ Thus}$$

$$\begin{aligned}
 P\{S > 10\} &\approx P\left\{Z > \frac{10-14}{\sqrt{559.72}}\right\} \\
 &\approx P\{Z > -0.1479\} \\
 &= P\{Z < 0.1479\} \\
 &= \Phi(0.1479) \\
 &\approx 0.5588.
 \end{aligned}$$